Singer Xing

Madison, WI
https://www.linkedin.com/in/mochi-xing-08390935b/
https://github.com/SingerXing | https://singerxing.github.io

EDUCATION

University of Wisconsin-Madison | Madison, WI **May 2027**
Bachelor of Arts: Data Science & Mathematics, GPA: 4.0/4.0

- Relevant Courses: Linear Algebra, Elementary Analysis, Analysis I(Honors), Elementary Topology, Ordinary Differential Equations(Honors), Numerical Analysis, Data Sci Programming I &II, Data Science Modeling I, Mtrx Mthds in Machine Learning, Stochastic Process, Linear Optimization, Introduction to Time Series, Data Science Modelling II, Partial Differential Equations

RELEVANT PROJECTS

Learning Statistically Accurate Dynamics of Complex Systems | **UW-Madison MXM** 02.2026 - 05.2026

- Built a NumPy/PyTorch autoencoder–LSTM baseline for stochastic Lorenz–96 dynamics, reducing 40 variables to a 13-dimensional representation and evaluating reconstruction and forecasts against reference trajectories.
- Investigated statistical-observation extensions using regional averages and Fourier features through moment augmentation and learned observation maps, with proposed EnKF corrections.

Data Website Development Project 02.2026 - 05.2026

- Developed a Flask web application to share datasets, display JSON-formatted data, and present visual insights through SVG plots.
- Implemented email validation and A/B testing to improve subscription functionality and donation-page performance.

Machine Learning Engineer Projects 09.2026 - present

- Built a Python decision-planning system using Bayesian updates, two-step lookahead, and shared setup-cost accounting to select inspections under budget and prerequisite constraints.
- Benchmarked five strategies across eight synthetic scenarios, demonstrating 14.6% lower mean modeled loss than no acquisition; expanded evaluation to 45 scenarios and verified the implementation with 197 automated tests.

INDUSTRY EXPERIENCE

STEALTH Ventures| Remote 08.2026 - 09.2026
Quantitative Analysis Volunteer

- Built an NLP classification pipeline with TF-IDF and LinearSVC; compared Grid, Random, and Bayesian hyperparameter optimization, improving test macro F1 from 0.7364 to 0.7505.
- Fine-tuned a transformer model on startup and investment text data, performing data cleaning, tokenization, train/validation/test splitting, and baseline-vs-fine-tuned performance analysis.

AWARDS & SCHOLARSHIPS

College of Letters & Science | University of Wisconsin Madison | WI 01.2025 - present

- 2025 Spring & 2025 Fall & 2026 Spring Dean's List
- Study Abroad Scholars
- Undergraduate Scholarships for Summer Study

Department of Mathematics Scholarships | University of Wisconsin Madison | WI

- R. Creighton Buck Prize

SKILLS & INTERESTS

Technical & Related Skills: R (Advanced), Excel (Advanced), Python (Advanced), SQL (Advanced), Mathematical Modelling, Web Scraping, Data Cleaning, Data Visualization, A/B Testing, Matplotlib, Dashboard Development
Interests: Yachting, Basketball, Skiing, Handcrafting